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Database Programming with PL/SQL

14-2

Understanding Remote Dependencies

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Objectives

- This lesson covers the following objectives:
 - Describe remote dependencies
 - List how remote dependencies are controlled
 - Describe when a remote dependency is unsuccessfully recompiled
 - Describe when a remote dependency is successfully recompiled

You CANNOT create a remote dependency in the Academy Application Express environment because there is only one database. In real life, you (or the DBA) would first create a database link, which is a pointer to a remote database, and then reference the database link within a local procedure.

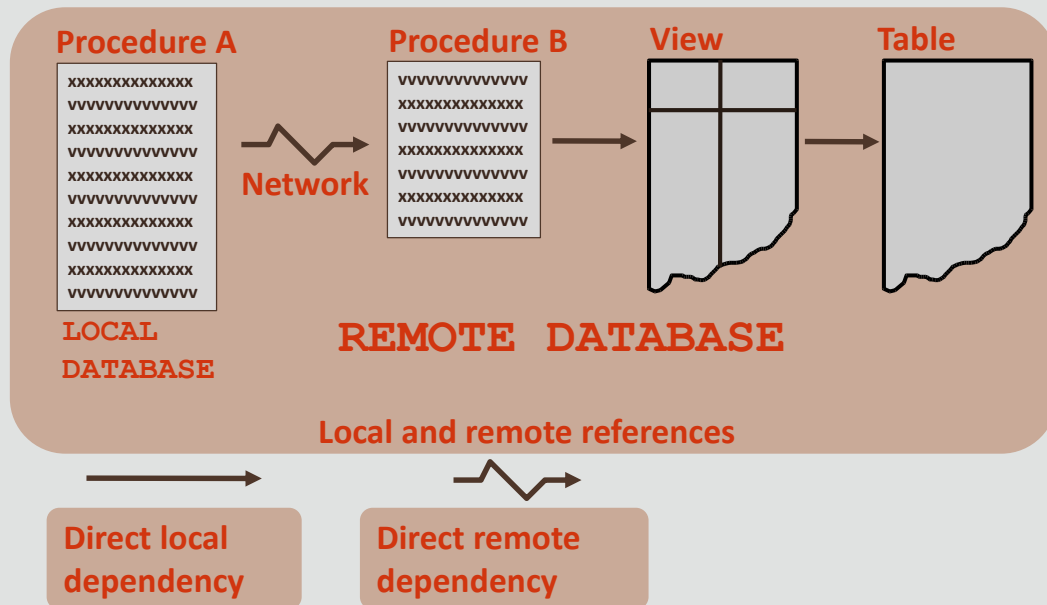
Purpose

- You have already learned that when a referenced object and a dependent object are in the same database, the Oracle server automatically records their dependency in the Data Dictionary
- But if the two objects are in different databases, this dependency between them is not automatically recorded
- Each database has its own Data Dictionary, so the status of the two objects is recorded in two separate Data Dictionaries, which do not automatically update each other across the network

Understanding Remote Dependencies

- When the referenced and dependent objects are in two different databases, the dependency is called a remote dependency
- The Oracle server does not automatically track relationships (such as dependencies) between objects in two different databases, even when the two databases are on the same server machine
- This is because each database has its own separate and independent data dictionary

Understanding Remote Dependencies

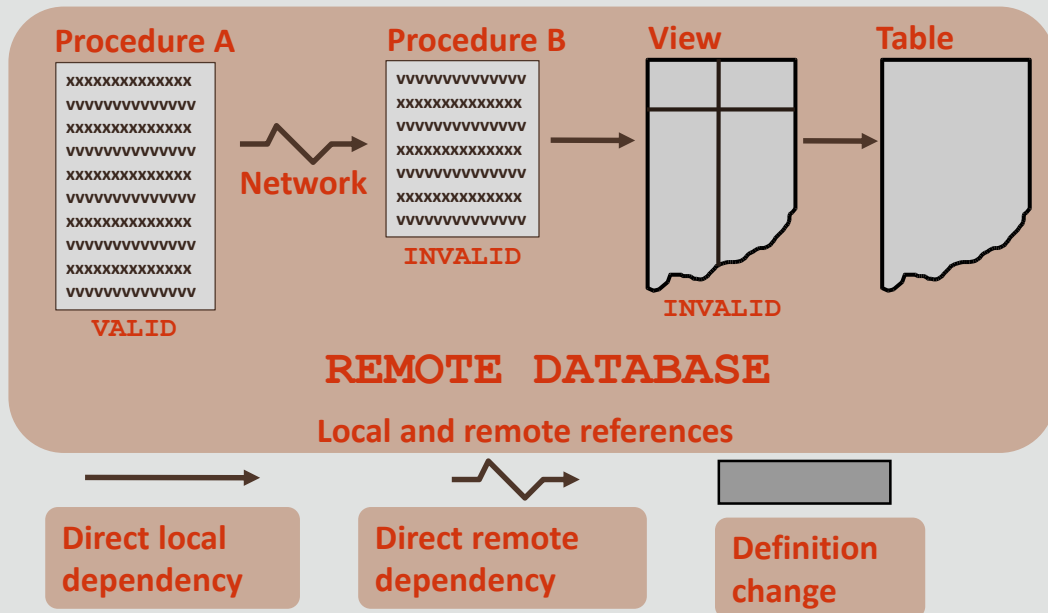


Understanding Remote Dependencies

- You CANNOT create a remote dependency in the Academy Application Express environment because there is only one database
- In real life, you (or the DBA) would first create a database link, which is a pointer to a remote database, and then reference the database link within a local procedure
- For example:

```
CREATE DATABASE LINK my_db_link
CONNECT TO remote_username IDENTIFIED BY remote_password
USING 'remote_database_name';
CREATE OR REPLACE procedureA ...
IS BEGIN
... procedureB@my_db_link (parameters...)...END;
```

Understanding Remote Dependencies



Understanding Remote Dependencies

- If the remote procedure is (directly or indirectly) invalidated, the server cannot automatically invalidate the local procedure because there is no reference to the local procedure within the remote data dictionary
- The previous slide shows what happens when a table in the remote database is modified
- The view and procedure in the remote database are automatically invalidated because they are in the same database as the table, but the local procedure is not invalidated because it is in a different database
- However, the local procedure can (and will) be invalidated when it invokes (calls) the remote procedure; in other words, at execution time

How are Remote Dependencies Managed?

- Because remote dependencies are not automatically tracked in data dictionaries, there must be another way for the Oracle server to check if a remote procedure has been invalidated
- There are two ways of doing this:
 - TIMESTAMP mode checking (the default)
 - SIGNATURE mode checking
- To use signature mode, set the local database parameter:
 - `REMOTE_DEPENDENCIES_MODE = SIGNATURE`

How to Change the Dependency Mode

- You can alter the dependency mode for the current session by using the ALTER SESSION command:

```
ALTER SESSION SET REMOTE_DEPENDENCIES_MODE =  
{SIGNATURE | TIMESTAMP}
```

- And, you can alter the dependency mode system-wide after startup with the ALTER SYSTEM command:

```
ALTER SYSTEM SET REMOTE_DEPENDENCIES_MODE =  
{SIGNATURE | TIMESTAMP}
```



What is Timestamp Mode?

- Whenever any database object is created or modified (for example when a PL/SQL procedure is recompiled), Oracle automatically records the timestamp of the change in the data dictionary
- You can see these timestamps in the USER_OBJECTS dictionary view
- For example, to see when your procedures were last compiled:

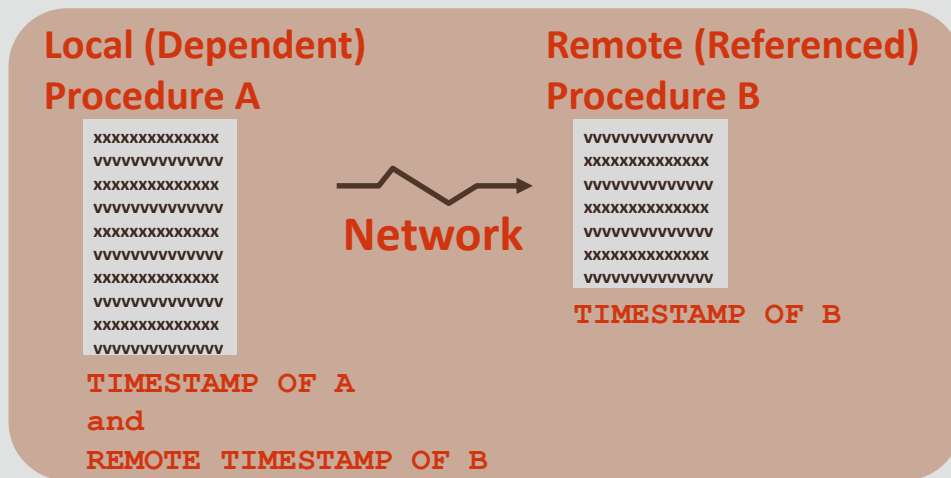
```
SELECT object_name, timestamp  
FROM USER_OBJECTS  
WHERE object_type = 'PROCEDURE';
```

These automatic timestamps are stored for all object types (table, view, index, synonym, and so on), not only for PL/SQL subprograms.

Timestamps and PL/SQL Subprograms

- When local procedure A is compiled, and it references remote procedure B, the remote procedure's timestamp is read and stored within the local procedure, as well as the local procedure's timestamp

Timestamps and PL/SQL Subprograms



Example of Timestamp Mode

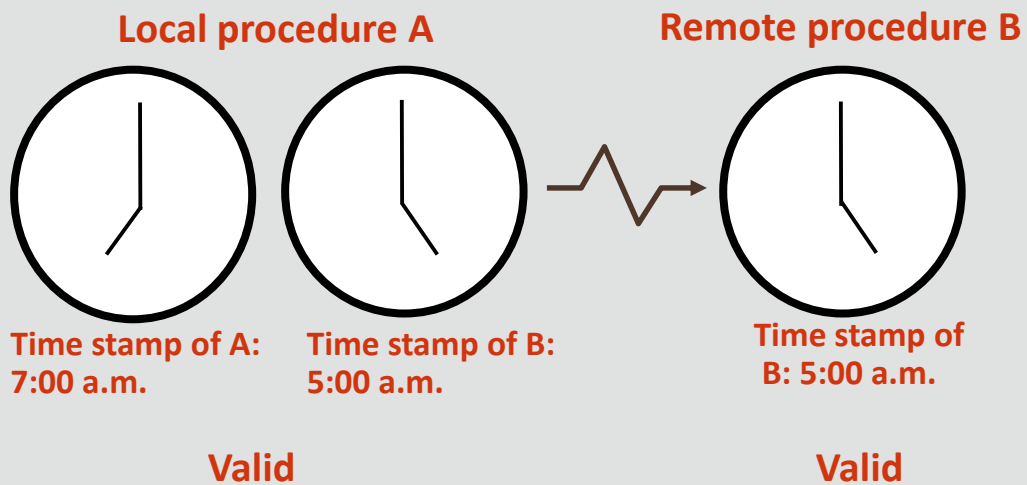
- Let's examine an example of how timestamp mode is used to check if a local subprogram is still valid, or must be invalidated
- To start with, imagine that both local procedure A and remote procedure B are valid
- Procedure B was compiled at 5:00 a.m., and procedure A was compiled at 7:00 a.m.



Example of Timestamp Mode

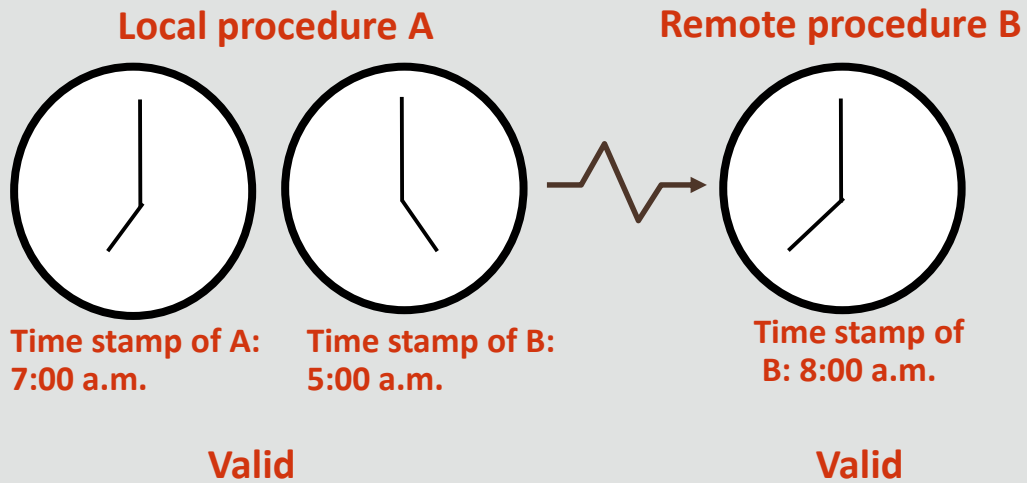
- When procedure A was compiled (at 7:00 a.m.) the compiler checked the validity and timestamp of remote procedure B by a call across the network, and stored B's timestamp within A's dictionary, as well as A's timestamp

Example of Timestamp Mode



Example of Timestamp Mode

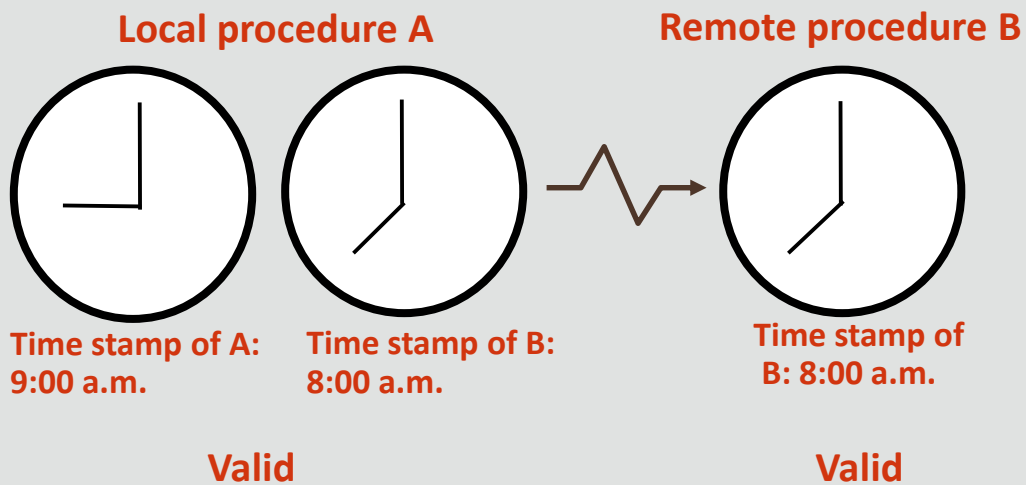
- Remote procedure B recompiles successfully at 8:00 a.m.



Remember that procedure A calls procedure B, but not vice-versa. So there is no reference to A in the code of B. Therefore, (re)compiling B stores B's new timestamp only.

Example of Timestamp Mode

- Local procedure A recompiles successfully at 9:00 a.m.



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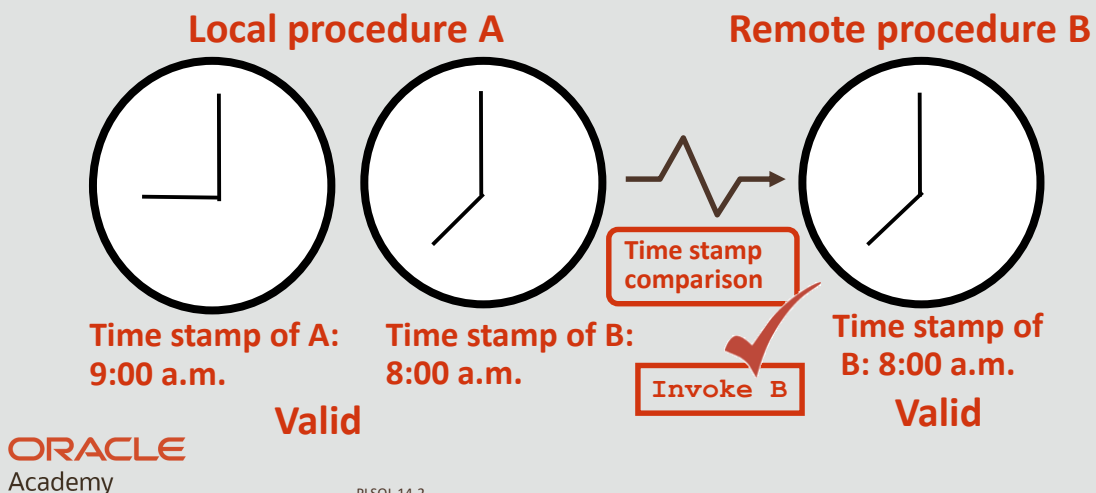
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The code of procedure A references procedure B. Therefore, when A is (re)compiled, the validity of B is checked, just as with local dependencies. The timestamp of B is stored within procedure A.

Example of Timestamp Mode

- Execute procedure A after 9:00 a.m.
- The two timestamps of B are compared and are equal, so A remains valid and executes successfully



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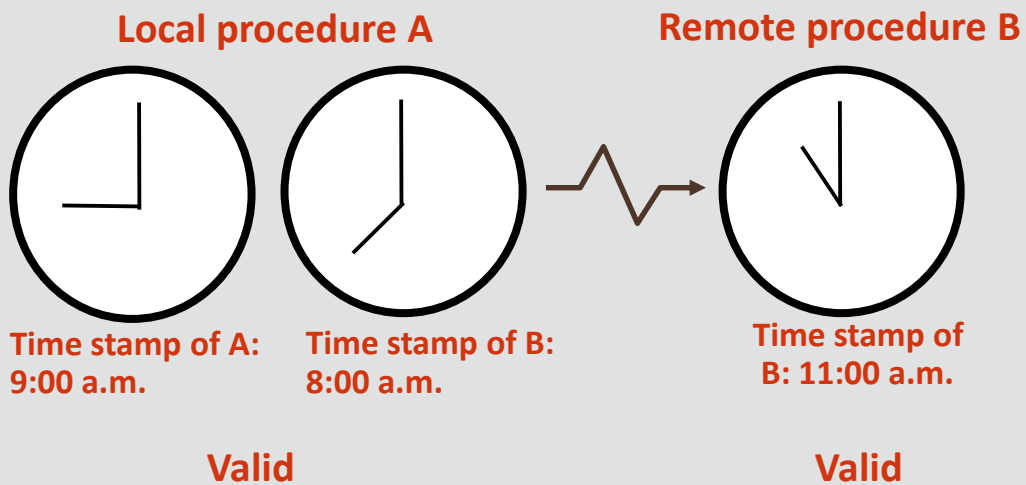
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When procedure A is executed, the validity and timestamp of B are checked. Because B is valid and its timestamp is still 8:00 a.m., the local Oracle server knows that B cannot possibly have been recompiled (therefore cannot have been altered) since 8:00 a.m.

Procedure A therefore remains valid and executes successfully.

Example of Timestamp Mode

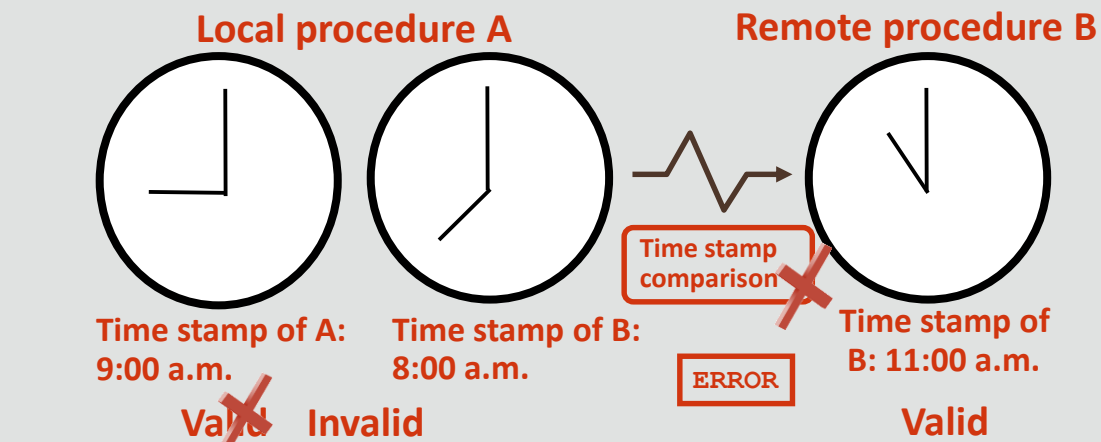
- Remote procedure B is recompiled successfully at 11:00 a.m.



This time, when procedure A is executed, the two timestamps of B are not equal. So the local Oracle server knows that B must have been recompiled since it was last invoked from A.

Example of Timestamp Mode

- Execute procedure A after 11:00 a.m.
- The two timestamps of B are not equal, so the execution fails and A is invalidated



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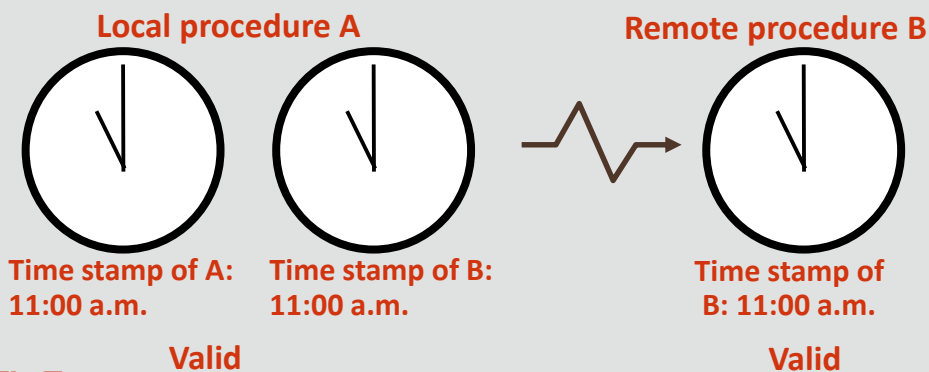
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The key point here is that A is not automatically recompiled, because it was not invalid at the start of its execution. Instead, the execution fails and returns an unhandled exception. This is true even when B's formal parameters were not changed by the recompilation at 11:00 a.m.

If the relationship had been a LOCAL dependency, and B's formal parameters had not changed, this step would have executed successfully and A would have remained valid.

Example of Timestamp Mode

- Execute procedure A again
- Because it was invalid, it is automatically recompiled and the new timestamps are stored
- This execution is successful



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The second execution of procedure A forces a recompilation because its status was invalid. As always, the actual parameters in A are compared with the formal parameters of B. Since they still match, A recompiles and executes successfully and the remote timestamp is updated.

B's new timestamp is stored within B.

Disadvantage of Timestamp Mode

- Did you notice that procedure A was marked invalid and its execution failed because procedure B's local and remote timestamps were not equal?
- This happened even though procedure B had recompiled successfully and was valid
- So remote dependencies sometimes cause unnecessary failures and later recompilations
- To avoid this limitation, we can use Signature Mode

Signature Mode

- In Signature Mode, a unique number called a signature is calculated and stored each time a procedure is recompiled
- The signature of a procedure is calculated from:
 - The name of the procedure
 - The data types of the parameters
 - The modes of the parameters
- So, the signature changes only if the procedure name or parameters are changed, not if a change is made to the body of the code

You cannot SELECT signatures from the data dictionary.

Signature Mode

- The signature of the remote procedure is saved in the local procedure, just like timestamp mode
- You cannot view the signature. It is not stored in USER_OBJECTS or any other Data Dictionary view
- Signature mode avoids unnecessary failures and recompilations, because the signature changes only when the dependent object would have to be recompiled anyway



Signature Mode Example

- REMOTE_DATABASE_NAME is a pointer to the remote database called a database link, which is not covered in this course

Remote at 8:00 a.m.>

```
CREATE OR REPLACE FUNCTION remote_func RETURN NUMBER IS
v_remote_count NUMBER;
BEGIN
SELECT COUNT(*) INTO v_remote_count FROM employees;
RETURN v_remote_count;
END remote_func;
```

Local at 9:00 a.m.>

```
CREATE OR REPLACE PROCEDURE local_proc IS
v_local_count NUMBER;
BEGIN
v_local_count := remote_func@remote_database_name;
END local_proc;
```

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Signature Mode Example

- The remote function has been recompiled with different code, but the function name and its parameters have not changed
- Therefore, its signature has not changed, and the local procedure will not be invalidated next time it is executed

Remote at 11:00 a.m.>

```
CREATE OR REPLACE FUNCTION remote_func RETURN NUMBER IS
v_remote_count  NUMBER;
BEGIN
SELECT COUNT(*) INTO v_remote_count FROM employees
WHERE salary > 10000;
RETURN v_remote_count;
END remote_func;
```

Signature Mode Example

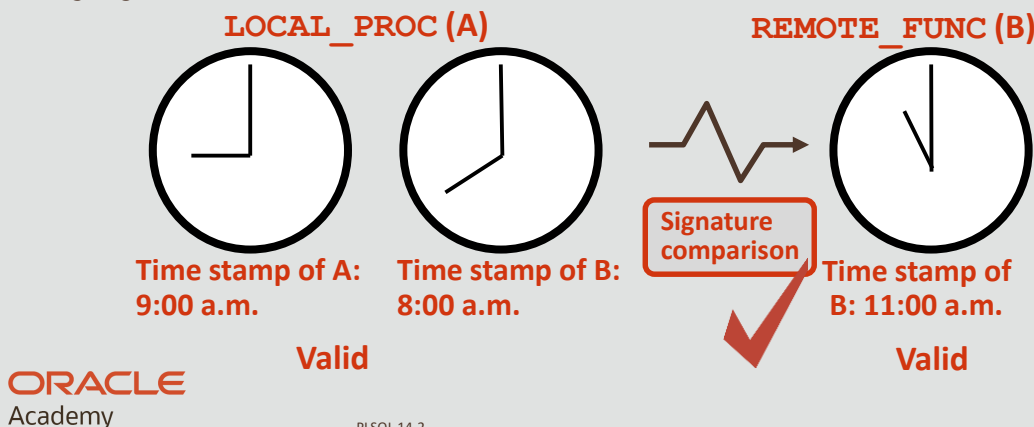
- Of course, if we changed something that affects the calling procedure (for example, changing the RETURN data type to BOOLEAN), the local procedure would still be invalidated

Remote at 11:00 a.m.>

```
CREATE OR REPLACE FUNCTION remote_func RETURN NUMBER IS
v_remote_count  NUMBER;
BEGIN
SELECT COUNT(*) INTO v_remote_count FROM employees
WHERE salary > 10000;
RETURN v_remote_count;
END remote_func;
```

How Signature Mode Works

- Execute LOCAL_PROC after 11:00 a.m.
- The two timestamps of REMOTE_FUNC are not equal, but their signatures are equal
- The execution succeeds and LOCAL_PROC remains valid



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Since the recompilation of procedure B at 11:00 a.m. did not change its formal parameters (and therefore they still match the actual parameters within procedure A), B's signature was not changed by its recompilation. Therefore, when A is executed, the local and remote signatures are still equal and the execution is successful.

Terminology

- Key terms used in this lesson included:
 - Remote dependency
 - Signature mode
 - Timestamp mode

- Remote Dependencies are when the Oracle database manages dependencies in a distributed environment across a network.
- Signature Mode is the setting to check if a local subprogram is still valid, or must be invalidated by checking the value a unique number calculated and stored each time a procedure is recompiled.
- Timestamp Mode is the setting to check if a local subprogram is still valid, or must be invalidated by checking the value in the USER_OBJECTS dictionary view.

Summary

- In this lesson, you should have learned how to:
 - Describe remote dependencies
 - List how remote dependencies are controlled
 - Describe when a remote dependency is unsuccessfully recompiled
 - Describe when a remote dependency is successfully recompiled

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